

Wout Vossen

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EXPERIENCE

Machine Learning Engineer (NLP & GenAI team)

Jan. 2024 – Present

CM — Belgium's largest health insurance fund (4.5M+ members)

Brussels, Belgium

- Architected CM's first GenAI framework powering **10+** production assistants and cross-team applications.
- Launched mail-reply AI assistants generating **15,000+** replies/day to incoming member emails.
- Established automated evaluations for AI assistants using classical NLP and LLM-as-judge metrics.
- Fine-tuned EmbeddingGemma to match Cohere Embed v4 on internal retrieval benchmarks.
- Implemented AWS SageMaker ML Pipelines for model training and serving, later adopted by sibling teams.
- Automated analysis of **10,000+** daily member interactions using AI-powered Dagster pipelines.
- Built CI/CD pipelines, reusable Terraform modules, and Grafana dashboards to deploy and monitor AI assistants on AWS.
- Led the AI Community with **100+** average attendees across **7** sessions to promote AI adoption across the division.

PROJECTS

Pookie — YouTube channel on applied AI | youtube.com/@pookiehd

Dec. 2024 – Present

- **7,000+** subscribers.
- Top video: "How to Finetune LLMs with Unsloth" — **70k+** views.
- Create approachable applied-AI content covering LLM fine-tuning, FlashAttention with Triton, local inference, diffusion models, and coding agents.

PsyScribe — AI mental-health assistant | *React, Flask, Google Cloud*

Jul. 2023 – Feb. 2024

- Commercialized my master's thesis into an AI mental-health product at pyscribe.com.
- Reached **30+** monthly paid subscribers.

crev — AI Code Reviews from the CLI | *Go, Google Cloud*

Aug. 2024 – Dec. 2024

- Built a CLI tool that bundles a codebase into a single file and generates AI-assisted code reviews.
- Reached **39** GitHub stars and **5** monthly paid subscribers.

PUBLICATIONS

Vossen, W., Szymanski, M., Verbert, K. *The effect of personalizing a psychotherapy conversational agent on therapeutic bond and usage intentions*. ACM IUI 2024, Greenville, SC — **35** citations.

Sileo, D., Vossen, W., Raymaekers, R. *Zero-Shot Recommendation as Language Modeling*. ECIR 2022 — **92** citations. Task contributed to Google BIG-bench, the foundational LLM evaluation benchmark used across frontier AI labs.

EDUCATION

KU Leuven

Sept. 2021 – Jul. 2023

Master of Engineering, Computer Science (option AI) — magna cum laude

Leuven, Belgium

KU Leuven

Sept. 2018 – Jul. 2021

Bachelor of Science, Computer Science — cum laude

Leuven, Belgium

KU Leuven

Sept. 2015 – Jul. 2018

Bachelor of Science, Psychology — cum laude

Leuven, Belgium

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, SQL

ML / GenAI: PyTorch, Hugging Face, Sentence-Transformers, Unsloth, Agentic systems, RAG, Weights & Biases, LLM evaluation, llama.cpp

Cloud / Infra: AWS, ECS, FastAPI, Docker, Dagster, Terraform, Kubernetes, Jenkins, Grafana, Linux